1. Introduction to Microsoft Excel

Microsoft Excel is a spreadsheet application used to store, organise, calculate, analyse, and present data in a structured format.

Uses of Excel

Excel can be used for a variety of tasks, including the following:

- Maintaining student records and attendance.
- Preparing budgets and salary reports.
- Performing calculations automatically.
- Analysing business and sales data.
- Creating charts, tables, and reports.
- Managing large datasets.

Example: A company can use Excel to maintain employee names, job titles, salaries, bonuses, and total compensation.

2. Workbook and Worksheet

Term	Meaning	Example
Workbook	An Excel file containing one or more worksheets.	Employee.xlsx
Worksheet	A single spreadsheet inside a workbook.	Data, Sheet1
Row	A horizontal arrangement of cells, identified by numbers.	Row 5
Column	A vertical arrangement of cells, identified by letters.	Column B
Cell	The intersection of a row and a column.	B5
Cell Range	A group of selected cells.	A1

Remember:

- Rows are horizontal and numbered 1, 2, 3, …
- Columns are vertical and lettered A, B, C, …
- A cell address consists of a column letter followed by a row number.
- The default Excel worksheet has 1,048,576 rows and 16,384 columns (A to XFD).

Example: Cell C5 is located at the intersection of Column C and Row 5.

3. Understanding the Excel Interface

Key Components of the Excel Interface

The Excel interface consists of several components that help users navigate the workbook, enter and edit data, and work with worksheets.

Component	Purpose
Title Bar	Displays the workbook name and the application.
Quick Access Toolbar	Provides quick access to commands such as Save and Undo.
Ribbon	Contains the tabs and tools used to work in Excel.
Name Box	Displays the address of the active cell, such as J3.
Formula Bar	Displays or allows editing of the content or formula of the active cell.
Worksheet Area	Provides the grid where we enter and work with data.
Active Cell	Refers to the cell that is currently selected.
Row Headings	Numbers displayed on the left side.
Column Headings	Letters displayed at the top.
Sheet Tabs	Allow us to switch between worksheets.
Status Bar	Displays information such as Sum, Average, and Count for selected cells.

Important Ribbon Tabs:

- **Home:** Font, alignment, number formatting, and cell editing.
- **Insert:** Tables, charts, and PivotTables.
- **Formulas:** Functions and formula-related tools.
- **Data:** Sorting, filtering, and data validation.
- **Review:** Comments and proofreading tools.
- **View:** Worksheet display and zoom settings.

4. Entering and Editing Data

Excel cells can contain:

- **Text:** Employee, Data Analyst, Delhi.
- **Numbers:** 100, 2500, 120000.
- **Dates:** 23-10-2026.
- **Formulas:** =A1+B1

How to enter data:

1. Select a cell.
2. Type the required value.
3. Press Enter.

How to edit data:

1. Select the cell.
2. Press F2 to edit its contents, or click the Formula Bar.
3. Make the changes and press Enter.

How to clear cell contents:

1. Select the cell or range.
2. Press Delete.

Clearing contents does not normally remove the cell's formatting.

5. Understanding Formulas

A **formula** is an expression used to perform calculations or comparisons in Excel.

Basic rules:

1. Every formula starts with an equal sign (=).
2. Cell addresses can be used instead of typing numbers manually.
3. Press Enter to calculate the result.

Example dataset:

A — Annual Salary	**B — Bonus**
120000	10000
135000	12000

To calculate the total salary in Row 2, enter:

=A2+B2

Result: 130000

If the value in A2 changes, the formula result updates automatically.

6. Arithmetic Operators

Arithmetic operators perform mathematical calculations.

Operator	Name	Purpose	Example	Result
+	Addition	Adds values	=10+5	15
-	Subtraction	Subtracts values	=10-5	5
*	Multiplication	Multiplies values	=10*5	50
/	Division	Divides values	=10/5	2
%	Percentage	Converts a number into a percentage	=50%	0.5

Examples using the salary dataset:

Examples Using the Salary Dataset

Annual Salary is in **D3**, and Bonus Max is in **E3**.

Total Salary

To calculate the total salary, add the annual salary and the bonus maximum:

=D3+E3

Difference Between Salary and Bonus

To calculate the difference between the annual salary and the bonus maximum, subtract the bonus from the salary:

=D3-E3

Double the Annual Salary

To calculate double the annual salary, multiply the value in D3 by 2:

=D3*2

Bonus as a Percentage of Annual Salary

To calculate the bonus as a percentage of the annual salary, divide the bonus maximum by the annual salary:

=E3/D3

After entering the last formula, use **Home → Number → Percentage** to display the result as a percentage.

For example, **10000 ÷ 120000 = 8.33%**.

Memory trick:

- Total → +

- Difference → -
- Times → *
- Divide → /
- Power → ^

7. Comparison Operators

Comparison operators compare two values. They return either TRUE or FALSE.

Operator	Meaning	Example	Result
=	Equal to	=10=10	TRUE
>	Greater than	=10>5	TRUE
<	Less than	=5<10	TRUE
>=	Greater than or equal to	=10>=10	TRUE
<=	Less than or equal to	=5<=10	TRUE
<>	Not equal to	=10<>5	TRUE

Important: A formula begins with =. When comparing values, the comparison operator is also used inside the formula.

Memory trick:

- At most → <=
- At least → >=
- Greater than → >
- Less than → <
- Different → <>

8. Cell References

A **cell reference** is the address of a cell used in a formula. There are three main types of cell references: relative, absolute and mixed.

A. Relative Reference

A relative reference changes when a formula is copied to another cell.

Example:

=D3+E3

When the formula is copied one row down, it becomes:

=D4+E4

Use: Calculating the total salary for multiple employees.

B. Absolute Reference

An absolute reference remains fixed when a formula is copied. It uses **$** before the column letter and row number.

Example:

=D3*B1

When the formula is copied down, **B1** remains fixed.

Use: Applying a fixed bonus rate or tax rate stored in one cell.

Shortcut: Select a cell reference inside a formula and press **F4** to cycle through reference types in desktop Excel.

C. Mixed Reference

A mixed reference fixes either the column or the row, but not both.

Reference	Meaning
$A1	Column A remains fixed; the row can change.
A$1	Row 1 remains fixed; the column can change.

Quick Revision

- A1 → Both can change.
- A1 → Neither changes.
- $A1 → Column fixed.
- A$1 → Row fixed.

9. Basic Excel Functions

A **function** is a predefined formula that performs a specific calculation.

General syntax:

=FUNCTION_NAME(argument)

An argument is the value, cell reference, or range supplied to a function.

Common Functions

Function	Purpose	Example
SUM	Adds numbers	=SUM(A1:A5)
AVERAGE	Calculates the arithmetic mean	=AVERAGE(A1:A5)
COUNT	Counts cells containing numbers	=COUNT(A1:A5)
COUNTA	Counts non-empty cells	=COUNTA(A1:A5)
MAX	Returns the largest value	=MAX(A1:A5)
MIN	Returns the smallest value	=MIN(A1:A5)
IF	Returns a result based on a condition	=IF(A1>=50,"Pass","Fail")
TODAY	Returns the current date	=TODAY()
SQRT	Calculates the square root	=SQRT(A1)
CONCAT	Joins text from different cells	=CONCAT(A1,B1)

Understanding ranges:

A1:A5 means all cells from A1 through A5, including both endpoints.

Example:

If A1 contains 10, 20, 30, 40, and 50:

- =SUM(A1:A5) returns 150.
- =AVERAGE(A1:A5) returns 30.
- =COUNT(A1:A5) returns 5.
- =MAX(A1:A5) returns 50.
- =MIN(A1:A5) returns 10.

Understanding IF:

=IF(A1>=50,"Pass","Fail")

Excel checks whether A1 is greater than or equal to 50.

- If TRUE, the result is Pass.
- If FALSE, the result is Fail.

Common Excel Errors

Excel displays an error when a formula cannot calculate correctly.

Common Excel Errors

Error Type	Example Formula	Easy Explanation
#DIV/0!	=10/0	Excel cannot divide a number by zero.
#VALUE!	=10+"Hello"	Excel cannot add a number and text in this example.
#REF!	=#REF!	The formula contains an invalid cell reference.
#NAME?	=COUNTT(10,20)	Excel does not recognise the misspelt function name.
#N/A	=NA()	The value is not available.
#NUM!	=SQRT(-1)	Excel cannot calculate the real square root of a negative number.
#NULL!	=SUM(A1:A3 C1:C3)	The formula requests an intersection between ranges that do not overlap.

Easy Tricks to Remember

- **#DIV/0!** → Division problem.
- **#VALUE!** → Wrong type of value.
- **#REF!** → Invalid reference.
- **#NAME?** → Incorrect name or spelling.
- **#N/A** → Value not available.
- **#NUM!** → Number problem.
- **#NULL!** → Incorrect range intersection.

Practice Questions

Identify the error in each formula:

1. =20/0
2. =SUMM(10,20)
3. =SQRT(-25)
4. =NA()